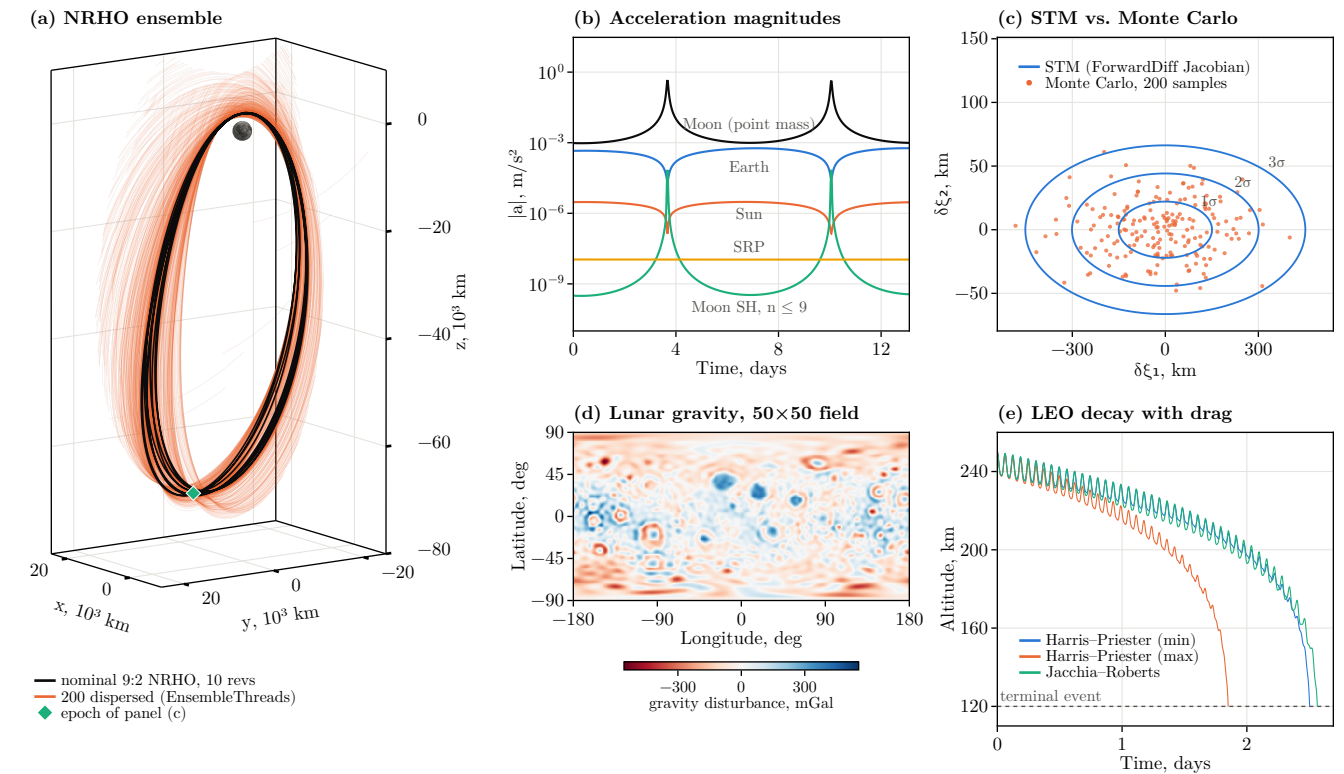

(a) NRHO ensemble
0
−20
−40
−60
−80
z, 10³ km
20
0
x, 10³ km
20
0
y, 10³ km
−20
nominal 9:2 NRHO, 10 revs
200 dispersed (EnsembleThreads)
epoch of panel (c)
(b) Acceleration magnitudes
10⁰
10⁻³
10⁻⁶
10⁻⁹
|a|, m/s²
Moon (point mass)
Earth
Sun
SRP
Moon SH, n ≤ 9
0
4
8
12
Time, days
(c) STM vs. Monte Carlo
STM (ForwardDiff Jacobian)
Monte Carlo, 200 samples
3σ
2σ
1σ
150
100
50
0
−50
δξ2, km
−300
0
300
δξ1, km
(d) Lunar gravity, 50×50 field
90
45
0
−45
−90
Latitude, deg
−180
−90
0
90
180
Longitude, deg
−300
0
300
gravity disturbance, mGal
(e) LEO decay with drag
240
200
160
120
Altitude, km
Harris–Priester (min)
Harris–Priester (max)
Jacchia–Roberts
terminal event
0
1
2
Time, days